

one set for input to the basic unit. Arrangement of pins is shown in Fig. 1.

IC1 is wired in the non-inverting comparator mode. The reference voltage is applied to inverting input (IN1-) and the voltage to be compared is applied to non-inverting input (IN1+). Whenever voltage to be compared (IN1+) goes above reference voltage (IN1-), output of the op-amp swings to high, and vice-versa. The alarm circuit is built around NE555. The frequency of the astable multivibrator is dependent on the values of resistors R4 and R5, and capacitor C2.

After connecting to 5V supply, wait for about ten seconds for the filament to heat properly. Adjust 10k potmeter (VR1) till the alarm stops sounding.

Under normal conditions (that is, no fumes from LPG or no alcohol in breath near MQ*), output of IC1 at pin 1 remains low. As a result, transistor T1 does not conduct, which

PARTS LIST

Semiconductors:

- IC1 - LM358 dual op-amp
- IC2 - NE555 timer
- T1 - BC547 npn transistor

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 10-kilo-ohm
- R2 - 4.7-kilo-ohm
- R3 - 220-ohm
- R4 - 1-kilo-ohm
- R5 - 56-kilo-ohm
- VR1 - 10-kilo-ohm potmeter

Capacitors:

- C1 - 100 μ F, 16V electrolytic
- C2 - 0.047 μ F ceramic disk
- C3 - 0.01 μ F ceramic disk
- C4 - 47 μ F, 16V electrolytic

Miscellaneous:

- CON1 - 2-pin connector terminal
- S1 - On/off switch
- LS1 - 8-ohm, 0.5W speaker
- MQ* - MQ2/MQ3/MQ6 sensor
- 5V regulated power supply
- 8-pin IC bases (2)

makes pin 4 of IC2 low. It keeps timer NE555 on reset. There is no output at pin 3 of IC2, and no sound comes out of the speaker.

On the other hand, when there are fumes from LPG, or there is al-

cohol in breath near MQ*, output of IC1 at pin 1 goes high. As a result, transistor T1 drives into saturation and its emitter goes high. Pin 4 of IC2 also goes high. The timer goes into astable mode, producing pulsed wave at output at pin 3 of IC2, which is coupled to speaker LS1 through coupling capacitor C4, and an alarm is set on.

Construction and testing

An actual-size, single-side PCB for the alarm is shown in Fig. 2 and its component layout in Fig. 3. Enclose the PCB in a suitable box, in such a way that fumes can circulate near MQ* sensor easily. Ensure proper wiring to avoid any errors. Before using the circuit, use MQ3 sensor to verify test points given in the table.

EFY note. The circuit was tested successfully with MQ3 alcohol sensor. ●

The author is B.Sc. (Physics) and a regular contributor to international magazines



Would You Like More Hands-on Circuits?

We Have **Thousands!**

VISIT TODAY

electronicsforu.com

If it's electronics, it's here